ROYAL CANADIAN NAVY

High Voltage Safety System of Work (SSOW) Authorized Engineer

QUALIFICATION STANDARD AND PLAN (QSP)

Marine Systems Engineering Officer (MSEO) (00345)
Marine Technician Electrican (MAR TECH) (00379-01)

Common Specialty Qualification – (AMBY)

MITE Course IDs:

Authorized Engineer (AE) – DL 123128

RECORD OF AMENDMENTS

This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear at the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J, “Course History.”

Revision	Approval Date	Authority
1.0 (QSPWB Month Year)		NTDC(P) CO
1.1	10 May 2024	CNTG

MARPAC RDIMS #1381614
Chap, 1, approved 10-05-2024
Chap 2, 3 approved 10-05-2024

TABLE OF CONTENTS

RECORD OF AMENDMENTS i
TABLE OF CONTENTS iii
FOREWORD iv
CHAPTER 1 TRAINING MANAGEMENT DETAILS 1
AIM 1
COURSE DURATION 2
RESOURCES 2
COURSE CAPACITY 3
MITE 3
COURSE REPORTING 3
PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR) 3
CHAPTER 2 ASSESSMENT OF TRAINEE 1
GENERAL 1
PERFORMANCE CHECKS 1
RESPONSIBILITY FOR TESTING 1
PROGRESS MONITORING 1
TRAINING REVIEW BOARD 2
COURSE REPORTS 2
CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS 1
ANNEX A – MAIN REFERENCES 15
ANNEX B – TASK LIST 1
ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS 1
ANNEX D – QSPWB SCALAR 1
ANNEX E – ON JOB PERFORMANCE RECORD (OJPR) 1
ANNEX F – ACRONYMS AND ABBREVIATIONS 1
ANNEX G – SUGGESTED SEQUENCING AND TIMINGS 1
ANNEX H – ADDITIONAL TRAINING RESOURCES 1
ANNEX I – COSTING INFORMATION 1
ANNEX J – COURSE HISTORY 1
ANNEX K – ASSESSMENT PLAN 1
ANNEX L – TEL PLAN 1
ANNEX M – TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS 1

FOREWORD

1 This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNTG) and in accordance with the following references:

a. A-P9 - 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification dated DD Month YY;

c. A-PD-055-002/PP-002 Non-Commissioned Member General Specification dated DD Month YY;

d. A-PD-055-002/PP-001 Name of Occupation Job Based Specification or Job-Based Occupational Specification dated DD Month YY;

e. A-PD-055-003/PQ-001 Name of Specialty Specification dated DD Month YY; and

f. NAVORD 4500-0 “Structured Delegation”; and

g. NPTORD 4500-22 “Qualification Standard and Plan Production and Management.”

2 This QSP consists of specific Performance Objectives (POs), Activity and Assessment Plan (AAP) in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3 This QSP is effective upon approval and supersedes previous versions.

Note: HV AOPV training is a new capability, and this is the first QSP version.

4 Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NFS(P) div), who will forward to NTG for approval, as appropriate.

5 The designated TEs for this training are not applicable; the HV interim training consists of eLearning via distributed learning and OJPR on ship.

TE	Phase / Module	MITE Course IDs	NQual

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

MARPAC RDIMS #1381614
Chap, 1, approved 10-05-2024
Chap 2, 3 approved 10-05-2024

AIM

1 The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the **Authorized Engineer (AE)**. This QSP is designed to assist the training system in developing, conducting, evaluating, and validating the training.

Authorized Engineer: AE qualified personnel are Naval Technical Officers as the Head of Department who are technically trained and knowledgeable with HV systems, HV SSOW, Canada Labour Code, and CSA electrical safety standards. In the absence of the NTO, the AE shall be the Senior Electrical Propulsion Manager. The AE is the only person who can recommend deviations from the HV SSOW to the Commanding Officer for approval.

Interim State Training Overview:

As part of the interim state training, the HV Individual Training & Education (IT&E) will be delivered through various delivery models based on HV qualifications:

- AE: Online instruction via DLN 3.0 as specified within Chapter 3.

COURSE DURATION

2 Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP. Note that the DL timings are approximations and will need to be validated during the post pilot course review.

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points	300 mins
Performance Checks (PCs)	50 mins per trainee + 5 min debrief
ATT Sub-total	
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	N/A
Training Administrative Time (TAT)	N/A
Training Travel Time (TTT)	N/A
Physical Fitness (PT)	N/A
Sub-total	N/A
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days (rounded up)**	
Total Training Days	N/A

RESOURCES

3 The following table details the key personnel required to conduct the training:

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
DLN 3.0 PC assessor	PO1 & above	Sr. Electrical Propulsion Mgr AE and AP Qualifications	1	PO 001, EOs 001.01-.03

4 The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP.

Critical Training Resources	PO/EO
1. Computer with internet access to DLN 3.0	PO 001, EOs 001.01-.03

COURSE CAPACITY

5 There shall be no course capacity limitation for DLN 3.0 online instruction. Planning for the conduct of training is based on the CFITES Vol 5 Annex D guidelines.

MITE

6 Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

7 Scheduling. Each course session shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System (NTS). It is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

8 Prerequisites. The loading authority must ensure that trainees have the following prerequisites before being selected for this course:

a. CSQ - High Voltage Authorized Person MITE 123127

9 Qualification. Successful completion of this course will grant the trainee the **High Voltage Authorized Engineer** qualification NQUAL AMBY and MITE 123128.

COURSE REPORTING

10 Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

11 PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces

Military Equivalencies Program", and CFITES Vol 12 "Canadian Forces Military Equivalencies Programme: Prior Learning Assessment".

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

MARPAC RDIMS #1337479
Chap, 1, approved DD MM YYYY
Chap 2, 3 approved DD MM YYYY

GENERAL

12 This is a pass/fail course. Trainees must successfully complete only the POs related to their occupation IAW NAVORD 11100-1 on ship represented in the following table:

Qualification	AE
Performance Objective	PO 001
Occupation	• NTO • SEPM (Mar Tech Elec)

13 Performance Checks (PCs) are based on the Performance Objectives (POs) in Chapter 3 of this QSP. Only the PCs related to the trainee's designated HV role must be successfully completed to achieve the qualification.

PERFORMANCE CHECKS

14 Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

15 A candidate unsuccessful in their first PC shall be permitted another PC rewrite. Upon unsuccessful completion of a second PC shall result in a recourse. The candidate may then redo the online instruction and challenge the PC based on the previously stated formula. No TRBs shall be conducted for unsuccessful PCs for this training.

RESPONSIBILITY FOR TESTING

16 Assessment of trainees is to be done IAW CFITES Vol. 7 "Evaluation of Leaners."

PROGRESS MONITORING

17 Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial

instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning so that content is retained.

18 Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes.”

19 Continuous monitoring of candidate progress is required to provide the following:

a. Early warning of trainee difficulties;
b. Feedback on the effectiveness of training; and
c. Information to assist in the completion of the CF 377 Course Report.

20 Instructional staff shall maintain individual trainee files, including the results of all:

a. Summative assessment material (PCs, CECs).

TRAINING REVIEW BOARD

21 There shall be no course capacity limitation for DLN 3.0 instruction.

COURSE REPORTS

22 Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51 “Course Reports and Certifications.”

23 Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide the career manager information.

24 The Course Report narrative comments could include:

a. Relevance of courseware;
b. Effectiveness of instruction; and
c. Effectiveness of assessment in measuring knowledge.

25 Training limitations and/or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

MARPAC RDIMS #1337479
Chap 1 approved DD MM YYYY
Chap 2, 3 approved DD MM YYYY

PO 001 – Complete HV SSOW Administration Role of an Authorized Engineer.

1. **Performance:** Complete HV SSOW Administration Role as an Authorized Engineer.

2. **Conditions:**

   a. **Given:**

      (1) Supervision; and

      (2) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Any condition on AOPV;

      (2) Day or night; and

      (3) At sea or alongside.

3. **Standard:** The authorized engineer shall complete HV SSOW administration role to include:

   a. Approving the authorization of HV work onboard AOPV;

   b. Updating the maintained switchboard register certificates onboard AOPV; and

   c. Performing HV safety procedure taskings as an Authorized Engineer onboard AOPV;

4. **References:** A1, A2, A3, A4, A5, A6, A7, C1 , C2.

5. **Task Numbers:** A7.

6. **Remarks:**

   a. Primary training for knowledge shall be available on DLN 3.0;

   b. Candidates must be qualified as an authorized person (AP) prior to obtaining the AE qualification.

EO 001.01 – Complete HV SSOV Administration Role of an Authorized Engineer

1. **Performance:** Complete HV SSOW Administration Role as an Authorized Engineer.

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall approve the authorization of HV work onboard AOPV IAW refs by:

   a. Authorizing all High Voltage Switching Schedule (HVSS) work by:

      (1) Visually checking the HVSS form is complete; and

      (2) Initialling step 1 of HVSS form.

   b. Authorizing all HVSP work by:

      (1) Visually checking HVSP forms are complete;

      (2) Signing HVSP forms to include:

         (a) PTW IAW SOP-S13C; and

         (b) SFT IAW SOP-S13D.

4. **Time Required:**

   a. EO – 2 x 50 min periods.

5. **Instructional Methods/Media:**

   a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.01 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A3, A4, A5, A6, A7, C1 , C2

9. **Remarks:**

a. Instruction is provided online because this PO is primarily focused as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Authorizing all High Voltage Switching Schedule (HVSS			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Initialling step 1 of HVSS form to include: • All relevant fields are initialled by NTO or SEPM (in NTO's absence)	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, High Voltage Switching Schedule Form
Visually checking the HVSS form is complete to include: • All fields are followed and initialled • A second AP has verified the form is complete	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, High Voltage Switching Schedule Form, Permit to Work Form, Sanction for Test Form

• Authorizing all HVSP work by: • Visually checking HVSP forms are complete • Signing HVSP forms to include: a) PTW IAW SOP-S13C; b) SFT IAW SOP-S13D	Direct instruction, EDI	Online delivery via DLN 3.0	

- EO 001.01 shall be assessed in PC 001

EO 001.02 – Complete HV SSOV Administration Role of an Authorized Engineer

1. **Performance:** Complete HV SSOV Administration Role of an Authorized Engineer

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall update the maintained switchboard register certificates onboard AOPV IAW refs by:

   a. Managing HV training for Naval Technical Department (NTD) by:

      (1) Monitoring HV training expiry dates in the certificate of appointment forms; and

      (2) Loading personnel on HV training courses;

   b. Conducting a certificate of appointment to include:

      (1) AE IAW Annex SOP-S13I;

      (2) AP IAW Annex SOP-S13J; and

      (3) CP IAW Annex SOP-S13K;

   c. Visually inspecting HVSP PPE and testing equipment maintenance records periodically;

   d. Conducting a monthly review of switchboard register to include:

      (1) Visually inspecting volumes 1 and 2;

      (2) Signing in volume 1 and 2 binders; and

      (3) Forwarding volumes 1 and 2 to CO for quarterly review and signature;

e. Conducting a monthly review of switchboard register to include:

(1) Visually inspecting volumes 1 and 2;

(2) Signing in volume 1 and 2 binders; and

(3) Forwarding volumes 1 and 2 to CO for quarterly review and signature;

f. Approving the master key register by:

(1) Visually checking SOP-S13M form is complete; and

(2) Initialling the SOP-S13M form.

4. **Time Required:**

a. EO – 3 x 50 min periods.

5. **Instructional Methods/Media:**

a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access; and

e. Computer with internet access.

7. **Test Details:**

a. EO 001.02 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A3, A4

9. **Remarks:**

a. Instruction is provided online because this PO is primarily focused on as an introduction for personnel authorizing work within HV spaces.

10. **Teaching Points:**

Lesson Objective: Update the maintained switchboard register certificates

Environment: Online (DLN 3.0)			Time: 3 x 50 min
Teaching Point	Method & Activities	Media & Resources	Refs
• Managing HV training for Naval Technical Department (NTD) by: a) Monitoring HV training expiry dates in the certificate of appointment forms; b) Loading personnel on HV training courses.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4
• Conducting a certificate of appointment to include: a) AE IAW Annex SOP-S13I b) AP IAW Annex SOP-S13J c) CP IAW Annex SOP-S13K.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Certificate of appointment Form
• Visually inspecting HVSP PPE and testing equipment maintenance records periodically	Direct instruction, EDI	Online delivery via DLN 3.0	A1, A2, A3, A4
• Conducting a monthly review of switchboard register to include: a) Visually inspecting volumes 1 and 2; b) Signing in volume 1 and 2 binders; and c) Forwarding volumes 1 and 2 to CO for quarterly	Direct instruction, EDI	Online delivery via DLN 3.0	A1, A2, A3, A4, Switchboard Register

review and signature.			
• Authorizing SEPM and APC handover certificates to include: a) Visually checking handover certificate forms; and b) Signing part 3 certificate of transfer.	Direct instruction, EDI	Online delivery via DLN 3.0	A1, A2, A4, SEPM Handover certificate, APC Handover certificate
• Approving the master key register by: a) Visually checking SOP-S13M form is complete; and b) Initialling the SOP-S13M form.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Master key register Form

- EO 001.02 shall be assessed in PC 001

EO 001.03 – Complete HV SSOV Administration Role of an Authorized Engineer

1. **Performance:** Complete HV SSOW Administration Role as an Authorized Engineer.

2. **Conditions:**

   a. **Given:**

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References;

   b. **Denied:** Nil.

   c. **Environment:**

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall perform HV safety procedure taskings as an Authorized Engineer onboard AOPV IAW refs by:

   a. Conducting periodic on-site safety audits by:

      (1) Visiting an HV work site; and

      (2) Visually inspecting safety measures are being followed IAW HV SSOW;

   b. Obtaining permission from CO for the emergency removal of HVSP locking devices; and

   c. Recommending deviations or amendments to approved HV safety rules to include:

      (1) Conducting a risk analysis IAW C23-005-000/AG-002 and AGG-040-010/AG-012; and

      (2) Verbally briefing the captain on ship.

4. **Time Required:**

   a. EO – 1 x 50 min period; and

   b. PC – 1 x 50 min period.

5. **Instructional Methods/Media:**

   a. Online instruction (DLN 3.0).

6. **Training Aids/Resources Required:**

   a. Workstation;

   b. Workplace;

   c. References;

   d. DLN 3.0 access; and

   e. Computer with internet access.

7. **Test Details:**

   a. EO 001.03 shall be assessed within PC 001. See Annex K Assessment Plan for details.

8. **References:** A1, A2, A3, A4, A5, A6, A7, C1, C2

9. **Remarks:**

   a. Instruction is provided online due to this PO being primarily focused as an introduction for personnel performing HV safety procedures.

10. **Teaching Points:**

Lesson Objective: Authorizing all High Voltage Switching Schedule (HVSS			
Environment: Online			Time:
Teaching Point	Method & Activities	Media & Resources	Refs
• Conducting periodic on-site safety audits by: a) Visiting an HV work site; b) Visually inspecting safety measures to ensure that the APC is following HV safety procedures IAW HV SSOW.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2

• Recommending deviations or amendments to approved HV safety rules to include: a) Conducting a risk analysis IAW C23-005-000/AG-002 and AGG-040-010/AG-012; and b) Verbally briefing the captain on ship.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, A7, C1, C2
• Obtaining permission from CO for the emergency removal of HVSP locking devices.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A6, C1, C2

- EO 001.03 shall be assessed in PC 001

PC 001 – Authorized Engineer Assessment

Assessment Activity
Strategy: Direct
Media: DLN 3.0
Method: Online Assessment via DLN 3.0
Total Time: 50 mins PC per trainee 5 mins feedback trainee (as required/if requested)
Environment: Any environment, any time, any location with access to DLN 3.0
References: • NAVORD 11100-1 • HDW Class SEMS - SOP S13 – HV Switching Program • CSA Z462 Workplace Electrical Safety • DAOD 2007-1 - General Safety Program • Canada Labor Code Pt. 2 • General Safety Program – General Safety Standards, Volume 2 Chapter 17 - Electrical • HDW Class SSO AL 11 25 April 2020 • Standard for Electrical Safety in the Workplace 2018
Pass/Fail Criteria: 80% and trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]). If a trainee is unsuccessful on a SAT checklist item, the assessment will cease.
Impact: Re-test upon first unsuccessful PC Recourse upon second unsuccessful PC
Limitations / Remarks:

Learners shall be provided the opportunity to redo the course after a second unsuccessful PC without the need for a TRB. There shall be no limit to the number of recourses for this course.
Assessment Details
Input: • Candidate shall be provided multiple-choice and short answer questions (filling in sample forms) via DLN 3.0. • A SME shall assess the scoring for short answers.
Assessment Process: • Candidate write PC 001 on DLN 3.0 at the end of the course
Output: Upon completion of the assessment, the trainee will have demonstrated proficiency in completing the HV SSOV Administration Role of an Authorized Engineer
Feedback / Debrief: Individual trainee feedback shall be provided after the assessment (as required for short answer questions for PC)

Assessment Resources		
Instructor Resources: Multiple-Choice + Short answer question bank.	Learner Resources: • PC 001 Checklist • References	To be Developed: Multiple-Choice + Short answer question bank.
Continuous Improvement		
After the lesson is delivered, the instructor will use this section to add improvements, suggestions and comments.		

ANNEX A – MAIN REFERENCES

A = CAF or other official Canadian government publication
B = Official Allied Force or Allied government publication
C = Commercial publication
D = Other

Ref Code	Publication
A1	NAVORD 11100-1
A2	HDW Class SEMS - SOP S13 – HV Switching Program
A3	DAOD 2007-1 - General Safety Program
A4	HDW Class SSO A11
A5	Canada Labour Code Pt. 2
A6	General Safety Standards, Volume 2 Chapter 17 - Electrical
A7	Damage Control Instructions 03/20 Home Port Alongside
C1	CSA Z462 Workplace Electrical Safety 2021

ANNEX B – TASK LIST

Task No	Task Statement	Train / No Train	NT Reasoning	PO 001 - AE
A7	Conduct AE, APC, AP, CP certificate of appointment process	T		AE
A20	Conduct SEPM handover process	NT	Prior learning – same as APC handover to be trained in AP course	AE
A22	Visually checking HVSP forms	NT	Task not difficult perform	AE
A23	Visually inspecting HVSP PPE and testing equipment	NT	Task not difficult to perform	AE
A24	Conduct monthly review of switchboard register	NT	Task not difficult to perform	AE
A25	Conduct periodic on-site safety audit	NT	Prior learning – AP course	AE
A26	Approve master key register	NT	Task rarely performed	AE
A27	Verbally brief captain on ship of deviations to approved safety rules	NT	Prior learning	AE

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 11 Mar 2021

Location: Virtual via MS Teams

Board Members:

Position	Rank and name	Role / Unit	Dates
Chair	PO1 Aaron Reid	Sr. Electrical Propulsion Mgr on HMCS HDW	11 Mar 21
SME	PO1 Marvin Murphy	LOTO Coordinator, FMF Cape Scott	11 Mar 21
SME	PO2 Michael Lamanque	Sr. Mech Specialist on HMCS HDW	11 Mar 21
TDO	SLt Sebastian Seo	TDO, Training Dev Centre Pacific	11 Mar 21
Secretary	Ms. Marilyn Carey	Civilian	11 Mar 21

Board Objective: To finalize the HV Safe System of Work QSP

Decision/Event	Discussion / Detail
Day 1: March 11, 2021	
SMEs	Given high voltage is so new to the RCN, there are no real SMEs for HV in the RCN
No task list	No approved HV task list from AOPV PMO or Occupational Authority or updated MAR Tech JTAR (last updated in 2019). Board referenced SEMS SOP-S13 document to identify HV tasks based on reality of the job.
Exclude devising	PO 2 Lamanque commented that there might be some circumstances where an APC may need to devise a handwritten sketch outside of the sketches provided on the pre-authorized

Decision/Event	Discussion / Detail
sketches for HVSS in QSP	HVSS form. Discussion about need to include this in QSP. HVSS pre-approved forms to be verified through QA on Mary Brooke (Seo). Decision: No train due to infrequency of event, prior training
Exclude HV maintenance tasks on QSP.	Discussed whether performing non-HV maintenance tasks should be inserted into QSP PO 002 (AP). Reid: methods for maintenance not changing depending on HV equipment. Only change is locking out. Lamanque: Maintenance is done HV Dead. Seo: MarTechs have prior training on maintenance. Murphy: I concur.
	Lamanque: Assume HV Live state until proven dead. Should this be included? Reid: Yes, I agree
Exclude performing SFT insulation testing	For insulation testing, prior learning for MAR Tech elecs on smaller circuits – already know how to do this (Reid). In reality, APs have been supervising contractors – this is also what the contract stipulates with external contractors (Murphy). Exclude from QSP
Exclude performing maintenance during SFT.	PO 002 (6) Discussion – SFT while at Sea. Lamanque: May be a situation at sea where ship staff will conduct work under an SFT. Reid: Yes, SFT includes running the machinery for testing purposes, e.g. propulsion converters updates on recent sail. Seo: Did you get formal training through Calhoon? Reid: Everything I've learned has been on my own. That's what we do as electricians.
Remove "HV safe" from PO 002 d.(6)	Reid: Only APC is to make the informative pipe for "HV Safe" during a damage control event, but the caveat is Duty Engineer would need to make this pipe while alongside and the ship is HV Live. Murphy: This is only related to when the ship is HV Live.
Change "switchboard safe" to "switchboard register" PO 002 d.(7)	Decision to change to "switchboard register" instead of "switchboard safe" because there is no place to mount the document safe on the wall on the AOPV ship because of fuel tanks, etc. (Reid)
PO 002 e. (10)	Discussion of role of safety member. Part of electrical training to never work on live circuits alone. If you need to test something, you would have safety member with 10 ft safety cane in case of accident (Murphy). Action of safety member = NO TRAIN (Seo). "Leave this one with me" (Seo)

Decision/Event	Discussion / Detail
PO 002 e. (8)	Placing all HVSS keys in switchboard key safe moved from (10) to (8). Important to put in QSP (Reid).
PO 002 f. (a.iv)	Added “visually inspecting worksite PPE.” Worksite PPE, not necessarily Arc Flash rated PPE needed. (Reid)
PO 002 f. (4.c.)	Discussion of signing off on both copies of PTW, including PiC’s copy. (Murphy) Both need to be completed. Added f. (4.c.) that one completed copy of PTW needs to go into switchboard register.
PO 002 g. (2)	AP is always the PiC but they can either undertake or supervise – focus on supervising (Seo). Our standard for this QSP is supervising, not doing. (Seo) Confirmed by Reid.
PO 002 h. 2(b) Are you always cordoning off the area with caution tape? (Murphy)	Not all the time, spaces are so big – never did an LOA in MCR, all LOAs so far main machinery room. These spaces are massive, HV is in one small area. Only once cordon off (Reid). There should never be concurrent work going on (Murphy). Decision to train personnel to cordon off. (Seo) Confirmed by Reid.
PO 002 i.	Added “Update HV Stateboards” (Murphy)
PO 002 k.	Do we need to train ppl: Managing all HV equipment, devices and accessories as an APC to include: Visually inspecting all HVSP equipment, devices and accessories periodically IAW Equipment Inspection Guidelines to include: Cleanliness; Damage; and Security. Tracking HVSP equipment, devices and accessories accessed by personnel; and Ordering replacement HVSP equipment, devices and accessories. Decision: No. Already trained duty engineers. This is no different. (Reid, Lamanque) NO TRAIN – Prior learning
PO 002 l.	Conducting a SEPM handover IAW Annex SOP-S13G with the oncoming Senior Electrical Propulsion Manager No Train as SEPM handover is same as APC Handover. It would be AE level.

Decision/Event	Discussion / Detail
	Completing HV safety program dev and evaluation technique profile – task rarely performed?
PO 002 m.	Completing HV safety program development and evaluation technique profile form IAW Annex SOP-S13O Not sure if this needs to be trained. Personally, I will do it. (Reid). Seems to be an experiential learning piece (Lamanque) NO TRAIN – task is done rarely. No SME experience with this. (Seo)
PO 001-3a.	Remove "Visually inspect and don PPE" from CP training. I've never seen a situation where the CP needs to don the PPE (Reid).
PO 001-3	Remove printing HV forms from CP level. AP would print these forms.
PO 001-3	"Completing" HVSS changed to "Reviewing" HVSS (Murphy).
PO 001-3	Conducting HV work on dead HV equipment to include: Changing air filter; Changing technical water; Changing oil; and Changing light bulb No train since MarTech Mechs already have prior training on CM and PM. (Reid, Lamanque, Murphy)
PO 001-3	Remove "Supervising work performed by HV aware personnel to include: a. Work being completed IAW LOA; b. Safety measures being followed IAW HV SSOW; and c. Work area is clean upon completion of work.
PO 003-3a,b	Train (a) based on SME response. AE must know the process. (Reid, Murphy) (b) These are different processes – all separate certificates, handwritten (Reid). CP Certification: WEng CANNOT work on any HV equipment; Mec's can only work on DEAD HV equipment; AP's can work on LIVE HV equipment– long process, must train (Reid).
PO 003-3c,d	Authorizing all High Voltage Switching Schedule (HVSS) work by: Visually checking the HVSS form is complete; Initialing step 1 of HVSS form. Authorizing all HVSP work by:

Decision/Event	Discussion / Detail
	Visually checking HVSP forms are complete; Signing HVSP forms to include: (a) PTW; SFT; and HVSS No Train (c) – leaning on APC, AE needs to know what work is being done and rationale behind it (Reid). 90% of work, they know it's coming (Murphy). Already trained in AP module (Seo). No Train (d) – Already trained in AP course. (Seo) Confirmed by Reid, Lamanque and Murphy.
PO 003-3e,f	Visually inspecting HVSP PPE and testing equipment maintenance records periodically; Conducting monthly review of switchboard register to include: (2) Visually inspecting volumes 1 and 2; Signing in volume 1 and 2 binders; and Forwarding volumes 1 and 2 to CO for quarterly review and signature. No Train (e) – just making sure records are being kept (Reid, Lamanque) Records not signed off – nothing to train, but need to be aware (Murphy). These should fall under "Responsibilities" for AE training (Lamanque, Seo) No Train (f) – easy task (Seo), knowledge base (Reid). Navord indicates CO signs monthly. (Murphy)
PO 003-3g, h	Conducting periodic on-site safety audits by: (3) Visiting an HV work site; Visually inspecting safety measures are being followed. Responding to HV near miss incidents to include: (4) Completing HV Near Miss Report SOP-S13P; Handing-over report to Unit General Safety Officer; Placing copy in switchboard register binder. NO Train (g) – prior learning in AP (Lamanque) NO Train (h) – admin (Lamanque), UGSO generates report (Murphy) simple task (Seo) AE Half day course, knowledge based on Sultan with test at the end (Reid)
PO 003-3 i,j	Obtaining permission from CO for the emergency removal of HVSP locking devices.

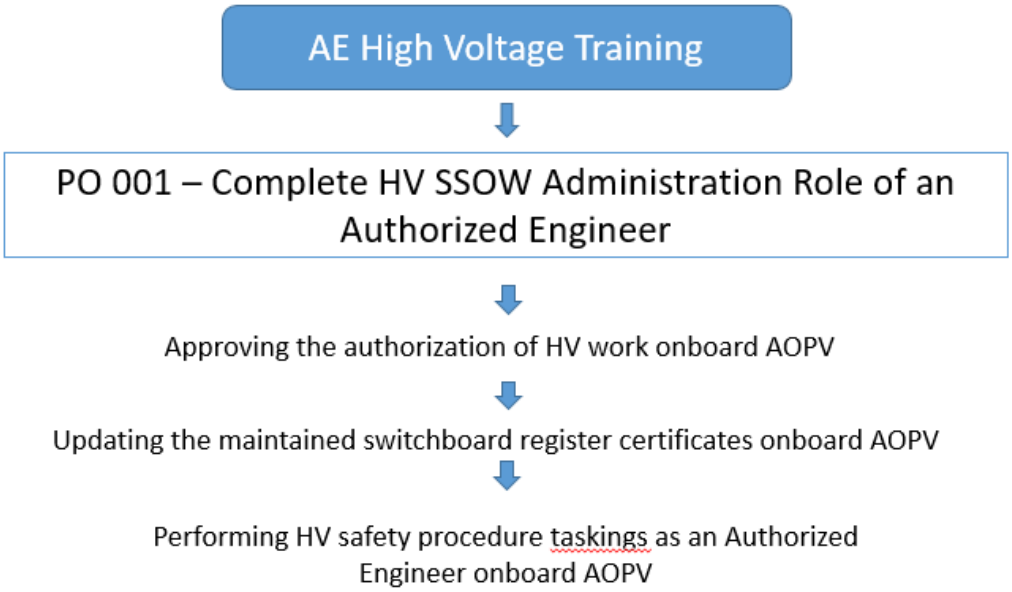
Decision/Event	Discussion / Detail
	Authorizing SEPM and APC handover certificate to include: d. Visually checking handover certificate forms to include: e. Senior Electrical Propulsion Manager handover certificate SOP-S13G f. APC handover certificate SOP-S13H Signing part 3 certificate of transfer No Train (i) simple task and/or not done rarely. Covered in LOTO system (Reid) No Train (j) SME's agreed.
PO 003-3 k,l	Approving the master key register by: g. Visually checking SOP-S13M form is complete; and h. Initialing the SOP-S13M form. i. Verbally briefing the captain on ship of deviations to approved safety rules. No train (k) – simple task, rarely done. Keys will never change b/c we use same keys and locks all the time. Takes awhile to set up. HDW every key and lock has one place to go. No train (l) – Already have training on how to talk to CO (Reid, Lamanque, Murphy)
AE Assessment?	Test should ask for references and where to find them (Murphy). Reid agreed. Scenarios? No need to request HMS Sultan Test – not relevant to AOPV (Reid)
Knowledge of references?	SEMS S-13, CSA bits and pieces, Canada Labour Code bits and pieces, Discuss later to figure out which pieces.
Day 2: March 17, 2021	
Assessment of PO 002	Regarding assessment, should the AP trainee be assessed on: (a) When closing the PTW, visually inspect work area, including clean work area, all tools and materials removed? (b) When closing the SFT, visually inspect work area, including clean work area, all tools and materials removed? (c) When closing the SFT, inform the AE and Duty Tech that work is complete, and DG 1 breaker is now available? Yes to a, b, c. These are all important duties for the APC. (Murphy, phone call with Marilyn). After the completion of a PTW or SFT, the PiC is responsible to ensure the worksite is

Decision/Event	Discussion / Detail
	cleaned up and the APC will inspect the worksite to ensure that it is complete before all final signatures. (Lamanque, email)

ANNEX D – QSPWB SCALAR

The mapping of the task analysis from the QSPWB is detailed in the standard paragraphs of the POs in Chapter 3 of this QSP, therefore, no scalar is inputted here.

The mapping of the instructional analysis from the QSPWB is detailed in the standard paragraphs of the EOs in Chapter 3 of this QSP, therefore, no scalar is inputted here.



ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

N/A for this QSP.

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym / Abbreviation	Term or Definition
AE	Authorized Engineer
AP	Authorized Person
APC	Authorized Person in Control
CME	Circuit Main Earth
CP	Competent Person
HV	High Voltage
HVSP	High Voltage Switching Program
HVSS	High Voltage Switching Schedule
LOA	Limitation of Access
NTD	Naval Technical Department
PiC	Person in Charge
POW	Point of Work
PPE	Personal Protective Equipment
PTW	Permit to Work
SEMS	Safety and Environmental Management System
SFT	Sanction for Test
SSO	Ship’s Standing Orders
SSOW	Safe System of Work
SEPM	Senior Electrical Propulsion Manager

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

Instructional time required – 300 mins
PC – 50 mins
Debrief – 5 mins

MARPAC RDIMS #1337479
Chap 1 approved DD MM YYYY
Chap 2, 3 approved DD MM YYYY

ANNEX H – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1 para 3-4, add any extra tables or paragraphs used for training resources in this annex. (Annex H not to be deleted)

ANNEX I – COSTING INFORMATION

Include information about the cost in whichever format is applicable (Not to be deleted).

Estimates only, based on CAF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

**5-year total based on one course per year, INCLUDING Initial Set-Up. Costs may be adjusted accordingly.

ANNEX J – COURSE HISTORY

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB (Dates)	(explanations of amendments, pertinent findings, trends, resource problems and modifications)
QSPWB (June 2022)	The dissection of the Authorized Engineer element of the QSP was to be separated into its single component.
Cont. Impr. 10 May 2024	Included prerequisite of AP to obtain AE.

ANNEX K – ASSESSMENT PLAN

OJPR/ PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
PC 001	PO 001	Demonstrate proficiency in the administrative role in authorizing HV work	Successful completion of online PC via DLN 3.0	80% Pass	Retry upon unsuccessful PC, Recourse upon unsuccessful 2nd PC.

ANNEX L – TEL PLAN

PO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta
PO 001	Online instruction via DLN 3.0	In-person instruction (if no TEL is available)	Lack of ability for learners to complete training due to limited qualified instructors and personnel.

ANNEX M – TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS

The following information was compiled during the QSPWB (date) to give context for developers. It can be removed from the QSP if desired after initial course development.

Key Target Population Details - (Course name) candidates

“The learners' level of skill and knowledge concerning the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners” -CFITES Vol 4

Characteristic	Target population details
General data	
Potential group size (based on the new AOPV ship): • AE: 2 (includes SEPM) for in-person instruction • Unlimited for DLN 3.0 online instruction Expected annual throughput: 25 based on untrained full AOPV NTD crew Location(s) of the intended learners: Currently in HFX for the first two AOPV ships and then ESQ once the third AOPV ship arrives west coast Access to computers (job): access to NFS computers and tablets Gender: Male & Female Age: 19 to early 50s Trades: MSEO, MAR Tech Elec	
Learning skills and preferences	
Competencies (general)	AE: Trained Engineer

Characteristic	Target population details
Reading and writing level	AE: Strong
Preferences, attitudes toward a specific type of learning material and equipment (i.e. TEL, aversion to lectures)	Open and flexible in using TEL
Educational level	University, College, High School
Previous military training experience	AE: HOD (8+yrs) AP: RQ-S1 qualified (4+ yrs)
Current subject-matter competence	
Experience with target performance	AE: None
Ability to perform any part of the POs	AE: Very High
Perception of the impact of achieving the POs on self, work or career	Excellent
Job/career stage	AE: mid-career (8+ yrs) AP: early to late-career (4+ yrs)